

The artifact I selected for this enhancement is my CS-330 3D Scene Project from Computational Graphics and Visualization. I originally created this project during CS-330 as a way to demonstrate concepts such as 3D object rendering, textures, lighting, transformations, and camera movement using C++, OpenGL, GLFW, GLEW, and GLSL shaders. The scene contains several objects, including a coffee mug, flashlight, dog tags, coin, and a textured tabletop. Users can also move around the scene using keyboard and mouse controls to view it from different angles. I chose this artifact because it represents one of the more challenging projects I completed during the Computer Science program. Most of my professional background has been in IT, working in positions such as IT Specialist, Network Administrator, and now Senior Desktop Support Analyst. While I have always enjoyed troubleshooting and supporting technology, projects like this helped me develop skills on the software development side and gave me an opportunity to work with concepts that were completely outside my normal day-to-day responsibilities.

I selected this project for the Algorithms and Data Structures category because it gave me a chance to improve how objects are stored and processed within the scene. In the original version of the project, each object was rendered individually using its own section of code. The project worked, but as I reviewed it for my code review and enhancement plan, I realized that adding more objects would eventually lead to a lot of repetitive code and make the project harder to expand. To address that, I created a custom SceneObject structure and used a vector to store object information such as textures, transformations, positions, and object types. Instead of handling every object separately, the program can now process multiple objects through iteration and a reusable rendering function. This allowed me to move away from repeatedly writing nearly identical rendering logic and toward a more scalable solution. What I like most about this

enhancement is that it demonstrates a practical use of data structures instead of simply implementing one for the sake of the assignment. The vector now serves a real purpose within the application, and the rendering loop allows objects to be processed more efficiently. While the project is still relatively small, the enhancement makes it much easier to add new objects in the future without creating entirely new rendering sections for each one.

This enhancement directly supports the course outcome involving the design and evaluation of computing solutions using algorithmic principles and computer science practices. Through the use of vectors, custom data structures, and iterative rendering, I was able to improve the overall design of the project while reducing repetitive code. It also reinforced the idea that choosing the right data structure can simplify future development and improve scalability. This enhancement also supports the outcome related to using industry-relevant tools and techniques. Working with OpenGL, C++, vectors, and object processing required me to apply concepts from multiple courses throughout the program. One thing I have noticed as I have progressed through the degree is that many of the concepts that originally felt separate now work together in larger projects. This enhancement was a good example of combining graphics programming with data structures to create a more flexible solution. My overall outcome plan remains the same. The previous milestone focused on software design and code organization, while this enhancement focused specifically on data structures, iteration, and object processing. My final enhancement will focus on the database category while continuing to avoid overlap between the three areas.

One of the biggest things I learned from this enhancement was how quickly repetitive code can build up, even in a project that is not very large. Looking back at the original version, it was easy to see that every new object required another block of rendering code. At first that seemed manageable, but it became obvious that the approach would not scale very well if the

project continued to grow. By creating a custom structure and using a vector to manage objects, I gained a better understanding of how data structures can help simplify repetitive operations and support future expansion.

The most challenging part of the enhancement was making changes without breaking the scene. Because every object depends on specific textures, transformations, and rendering settings, even a small mistake could cause objects to disappear or render incorrectly. There were several times during testing when objects were missing or materials were not displaying correctly, which required me to go back through the code and troubleshoot what changed. While it was frustrating at times, it also helped me gain a better understanding of how the rendering process works and how important testing becomes after even small code changes. Another challenge was making sure this enhancement stayed focused on algorithms and data structures rather than drifting back into software design and engineering. Since I am using the same artifact for all three enhancement categories, I had to be careful about separating the purpose of each milestone. Overall, this enhancement helped me better understand how data structures and algorithms can improve the flexibility of a program and reduce unnecessary repetition. It also reinforced the idea that writing code that works is important, but designing code that can grow and adapt is just as valuable.